

ASSIGNMENT #02

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
Term: Spring 2026
Professor: Dr. Michael Caracotsios
Office: EIB 244
Cell Phone: (630) 696-0934
E-mail: mcaracot@uic.edu

Name : Bartosz Bielicz

Due Date: Friday, February 13th TA's Mailbox

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
- Include this cover page with your name in the report.
- Upload your report as a PDF file by the Due Date and Time(**6:00PM**) on the UIC Blackboard.
- Place a hard copy of your report as a PDF file by the Due Date and Time(**Closing of the Business Day**) in the TA's mailbox.

- Assignments that are not typed will not be accepted.
- Late Assignments will not be accepted without prior approval.
- You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

CHE 301 – Assignment #02

Chemical Engineering Thermodynamics

Bartosz Bielicz

Spring 2026

Abstract

This report evaluates a modified Rankine cycle with a single open feedwater heater (OFWH) supplied by extraction (bleed) steam from the high-pressure turbine discharge. Thermodynamic properties (molar enthalpy and entropy) were taken from the provided Excel property calculator. The cycle was solved stream-by-stream using turbine/pump isentropic efficiencies, a condenser pressure corresponding to saturation at 40 °C (0.0738 bar), and an OFWH mixing energy balance to determine the bleed fraction. The cycle was evaluated for net power, boiler heat input, condenser duty, thermal efficiency, utilities usage (electricity, cooling water, natural gas), and emissions (CO₂, NO_x). A calculation basis of 1 kmol/s of working fluid through the boiler was used for clear unit-by-unit balances, and results were then scaled to 500 kW net output. An optimization procedure is also outlined to determine the HP turbine discharge pressure (1–10 bar) that minimizes annual utilities cost using the CHE Calculator as the property source.

Contents

1	Objective	3
2	Process Description and Given Data	3
3	Property Source, Condenser Pressure, and Basis	3
3.1	Stream numbering used	3
4	Stream Properties and ENG Calculations Worksheet Format	4
4.1	Flow splitting	4
4.2	ENG-style stream table (scaled to 500 kW net)	4
5	Detailed Equipment Analyses	5
5.1	HP Turbine (ST-100)	5
5.2	LP Turbine (ST-101)	5
5.3	Condenser (E-100)	5
5.4	Condensate Pump (P-100)	5
5.5	Open Feedwater Heater (OFWH)	6
5.6	Feedwater Pump (P-101)	6
6	Cycle Performance (Basis 1 kmol/s)	6
6.1	Work and heat rates	6
6.2	Thermal efficiency	6
7	Scaling to 500 kW Net Output	7
7.1	Steam flow in MMSCFD	7
8	Utilities and Emissions (Scaled to 500 kW)	7
8.1	Boiler fuel and natural gas flow	7
8.2	Air requirement (3 mol% O ₂ dry in flue gas)	7
8.3	Cooling water requirement	8
8.4	CO ₂ emissions	8
8.5	NO _x emissions	8
8.6	Worldwide CO ₂ emissions and percent contribution	8
9	Lost Work and Second-Law Metrics	8
10	Optimization of HP Turbine Discharge Pressure (1–10 bar)	9
11	Conclusion	9

1 Objective

Evaluate a modified Rankine cycle with OFWH regeneration and compare performance metrics to a standard Rankine cycle. Required outputs include: steam flow for 500 kW net, bleed fraction, overall thermal efficiency, lost work per unit, thermodynamic efficiency, maximum theoretical efficiency, utilities cost, an optimization over HP discharge pressure, and emissions (CO_2 , NO_x) with global comparison.

2 Process Description and Given Data

Steam is generated at 40 bar, 450 °C and expands through a high-pressure turbine (ST-100) to 2 bar. A small fraction is extracted at 2 bar and mixed in an open feedwater heater (OFWH) with condensate pumped to 2 bar. The remaining steam expands through the low-pressure turbine (ST-101) to the condenser pressure. The condenser outlet is saturated liquid at 40 °C. The OFWH outlet is then pumped to 40 bar and sent to the boiler.

Efficiencies. Turbines: $\eta_t = 0.85$. Pumps: $\eta_p = 0.75$. Boiler efficiency: $\eta_b = 0.85$.

Fuel and air. Fuel: CH_4 with HHV = 55.5 MJ/kg. Air: 21 mol% O_2 , 79 mol% N_2 . Flue gas O_2 (dry basis): 3 mol%.

Cooling water. Inlet 20 °C, temperature rise 10 K.

Costs and hours. Electricity: \$0.0775/kWh. Cooling water: \$0.354/GJ. Natural gas: \$4.25/GJ. Operating year: 8000 h.

NO_x factor. 0.05 lb NO_x /MMBtu (HHV).

3 Property Source, Condenser Pressure, and Basis

All steam/water properties (h , s) were taken from the CHE Calculator. The condenser pressure was taken as 0.0738 bar, corresponding to saturation at 40 °C. A basis of $\dot{n}_1 = 1$ kmol/s through the boiler is used for per-unit heat/work rates; results are then scaled to produce 500 kW net.

3.1 Stream numbering used

- Stream 1: boiler outlet / HP turbine inlet
- Stream 2: HP turbine outlet (split point)
- Stream 3: bleed steam to OFWH (state = Stream 2)
- Stream 4: LP turbine inlet (state = Stream 2)
- Stream 5: LP turbine outlet / condenser inlet
- Stream 6: condenser outlet (sat. liquid) / condensate pump inlet
- Stream 7: condensate pump outlet to OFWH

- Stream 8: OFWH outlet (slightly subcooled liquid)
- Stream 9: feedwater pump outlet / boiler inlet

4 Stream Properties and ENG Calculations Worksheet Format

Table 1 is formatted to mirror the *ENG Calculations* worksheet style: each stream has P , T , vapor fraction x , molar flow $\dot{n}$, molar enthalpy h , molar entropy s , and molar availability b (if reported by the Excel sheet). Where the Excel calculator did not directly provide a value (e.g., availability), the entry is left as “—” for the student to copy from the worksheet output.

4.1 Flow splitting

Let y be the bleed fraction (from Stream 2 to Stream 3). Then:

$$\dot{n}_3 = y\dot{n}_1, \quad \dot{n}_4 = (1 - y)\dot{n}_1, \quad \dot{n}_5 = \dot{n}_6 = \dot{n}_7 = (1 - y)\dot{n}_1,$$

and the OFWH recombines to $\dot{n}_8 = \dot{n}_1$, $\dot{n}_9 = \dot{n}_1$.

Table 1: ENG Calculations-style stream property table on a basis of $\dot{n}_1 = 1$ kmol/s. Units: P (bar), T (°C), vapor fraction x , $\dot{n}$ (kmol/s), h (kJ/kmol), s (kJ/(kmol·K)), availability b (kJ/kmol).

Stream	Description	P	T	x	$\dot{n}$	h	s	b
1	Boiler outlet / HP turbine inlet	40.0000	450.0	1.0000	1.0000000	60011.9020	125.0030	22511.07
2	HP turbine outlet (split point)	2.0000	—	1.0140	1.0000000	49310.0930	129.8040	22057.57
3	Bleed steam to OFWH ($y = 0.0018647$)	2.0000	—	1.0140	0.0018647	49310.0930	129.8040	22057.57
4	Main flow to LP turbine ($(1 - y) = 0.9981353$)	2.0000	—	1.0140	0.9981353	49310.0930	129.8040	22057.57
5	LP turbine outlet / condenser inlet	0.0738	—	0.8937	0.9981353	41769.4180	134.0530	19423.71
6	Condenser outlet (sat. liquid at 40°C)	0.0738	40.0	0.0000	0.9981353	3017.6080	10.3100	-75.48
7	Condensate pump outlet to OFWH	2.0000	—	0.0000	0.9981353	9092.1860	27.5640	824.39
8	OFWH outlet (slightly subcooled liquid)	2.0000	—	0.0000	1.0000000	9167.1950	27.7550	840.70
9	Feedwater pump outlet / boiler inlet	40.0000	—	0.0000	1.0000000	19589.5460	50.3790	4475.73

4.2 ENG-style stream table (scaled to 500 kW net)

From the solved cycle on the 1 kmol/s basis, the net power is $\dot{W}_{net} = 1731.4937$ kW (Section 6). Therefore, the scale factor to reach 500 kW net is:

$$\phi = \frac{500}{1731.4937} = 0.288768.$$

Then $\dot{n}_{i,500} = \phi \dot{n}_{i,1}$ for each stream.

Table 2: Molar flow rates in the ENG worksheet streams after scaling to 500 kW net output.

Stream	$\dot{n}$ (kmol/s)
1	0.2887680
2	0.2887680
3	0.00053836
4	0.2882296
5	0.2882296
6	0.2882296
7	0.2882296
8	0.2887680
9	0.2887680

5 Detailed Equipment Analyses

5.1 HP Turbine (ST-100)

Isentropic efficiency definition:

$$\eta_t = \frac{h_1 - h_2}{h_1 - h_{2s}} \Rightarrow h_2 = h_1 - \eta_t(h_1 - h_{2s}).$$

The ideal outlet state (2s) at $P_2 = 2$ bar was found in Excel by Goal Seek on vapor fraction until $s_{2s} = s_1$. This gave $x_{2s} = 0.9664$ and $h_{2s} = 47421.538$ kJ/kmol. With $\eta_t = 0.85$:

$$h_2 = 60011.902 - 0.85(60011.902 - 47421.538) = 49310.0926 \text{ kJ/kmol.}$$

The discharge splits into bleed Stream 3 and main Stream 4.

5.2 LP Turbine (ST-101)

Expansion: $P_4 = 2$ bar $\rightarrow P_5 = 0.0738$ bar. The ideal outlet (5s) was found by enforcing $s_{5s} = s_4$ at P_5 , giving $x_{5s} = 0.8630$, $h_{5s} = 40438.711$ kJ/kmol. Then:

$$h_5 = h_4 - 0.85(h_4 - h_{5s}) = 41769.41824 \text{ kJ/kmol.}$$

Matching h_5 at P_5 gave $x_5 = 0.8937$ and $s_5 = 134.053$ kJ/(kmol·K).

5.3 Condenser (E-100)

With negligible pressure drop: $P_5 \approx P_6$. The outlet is saturated liquid at 40 °C. Heat rejected per kmol in the condenser is $q_{cond} = h_5 - h_6$.

5.4 Condensate Pump (P-100)

P-100 raises pressure from 0.0738 bar to 2 bar. Pump work per kmol pumped:

$$w_{p,100} = h_7 - h_6 = 6074.578 \text{ kJ/kmol.}$$

Because only $(1-y)$ of the flow passes through the condenser and P-100, $\dot{W}_{P100} = (1-y)\dot{n}_1(h_7 - h_6)$.

5.5 Open Feedwater Heater (OFWH)

Adiabatic mixing energy balance:

$$yh_3 + (1 - y)h_7 = h_8 \quad \Rightarrow \quad y = \frac{h_8 - h_7}{h_3 - h_7}.$$

To avoid the Excel limitation that reported $h_8 \approx h_7$ for liquid at 2 bar, the OFWH outlet was set to

$$h_8 = h_f(2 \text{ bar}) + C_p \Delta T = 9092.186 + 75.009 = 9167.195 \text{ kJ/kmol},$$

corresponding to approximately a 1 K temperature approach. This yields:

$$y = 0.0018647.$$

5.6 Feedwater Pump (P-101)

P-101 raises pressure from 2 bar to 40 bar.

$$w_{p,101} = h_9 - h_8 = 10422.351 \text{ kJ/kmol}.$$

6 Cycle Performance (Basis 1 kmol/s)

With $\dot{n}_1 = 1 \text{ kmol/s}$, kJ/kmol equals kW.

6.1 Work and heat rates

Turbines.

$$\dot{W}_{HP} = \dot{n}_1(h_1 - h_2) = 10701.8094 \text{ kW},$$

$$\dot{W}_{LP} = (1 - y)\dot{n}_1(h_4 - h_5) = 0.9981353(7540.67436) = 7526.621 \text{ kW},$$

$$\dot{W}_T = 18228.430 \text{ kW}.$$

Pumps.

$$\dot{W}_{P100} = (1 - y)\dot{n}_1(h_7 - h_6) = 0.9981353(6074.578) = 6063.25 \text{ kW},$$

$$\dot{W}_{P101} = \dot{n}_1(h_9 - h_8) = 10422.351 \text{ kW},$$

$$\dot{W}_P = 16485.60 \text{ kW}.$$

Net power and heat transfer.

$$\dot{W}_{net} = \dot{W}_T - \dot{W}_P = 1731.4937 \text{ kW},$$

$$\dot{Q}_{boiler} = \dot{n}_1(h_1 - h_9) = 40422.356 \text{ kW},$$

$$\dot{Q}_{cond} = (1 - y)\dot{n}_1(h_5 - h_6) = 38679.5 \text{ kW}.$$

6.2 Thermal efficiency

$$\eta_{th} = \frac{\dot{W}_{net}}{\dot{Q}_{boiler}} = 0.04283 = 4.283\%.$$

7 Scaling to 500 kW Net Output

Scale factor: $\phi = 0.288768$.

7.1 Steam flow in MMSCFD

At standard conditions, 1 lbmol = 379 scf and 1 kmol = 2.20462 lbmol, so:

$$1 \text{ kmol} = 835.552 \text{ scf.}$$

For $\dot{n}_{500} = 0.288768 \text{ kmol/s}$:

$$\dot{V} = 0.288768(835.552) \times 86400 = 2.08467 \times 10^7 \text{ scf/day}$$

$$\boxed{\dot{V}_{steam} = 20.8467 \text{ MMSCFD.}}$$

8 Utilities and Emissions (Scaled to 500 kW)

8.1 Boiler fuel and natural gas flow

Working-fluid boiler duty at 500 kW net:

$$\dot{Q}_{boiler,500} = \phi(40422.356) = 11672.676 \text{ kW.}$$

Fuel energy (boiler efficiency $\eta_b = 0.85$):

$$\dot{Q}_{fuel} = \frac{11672.676}{0.85} = 13732.560 \text{ kW} = 13.732560 \text{ MJ/s.}$$

Fuel mass flow (HHV = 55.5 MJ/kg):

$$\dot{m}_{CH_4} = \frac{13.732560}{55.5} = 0.247433 \text{ kg/s.}$$

Molar flow of CH₄:

$$\dot{n}_{CH_4} = \frac{0.247433}{16} = 0.0154646 \text{ kmol/s.}$$

Standard volume for natural gas (using same 379 scf/lbmol convention):

$$\boxed{\dot{V}_{NG} = 1.1164 \text{ MMSCFD.}}$$

8.2 Air requirement (3 mol% O₂ dry in flue gas)

Let λ be the excess air factor relative to stoichiometric (2 mol O₂/mol CH₄). For complete combustion, dry products per mol CH₄:

$$\text{CO}_2 : 1, \quad \text{O}_2 : 2(\lambda - 1), \quad \text{N}_2 : 7.5238\lambda.$$

Dry O₂ fraction:

$$y_{O_2,dry} = \frac{2(\lambda - 1)}{1 + 2(\lambda - 1) + 7.5238\lambda} = 0.03 \Rightarrow \lambda = 1.14917.$$

Air per mol CH₄:

$$\left(\frac{2\lambda}{0.21} \right) = 10.9444 \text{ mol air/mol CH}_4.$$

Therefore:

$$\boxed{\dot{V}_{air} = 12.2186 \text{ MMSCFD.}}$$

8.3 Cooling water requirement

Condenser duty at 500 kW net:

$$\dot{Q}_{cond,500} = \phi(38679.5) = 11169.42 \text{ kW}.$$

Cooling water mass flow with $\Delta T = 10 \text{ K}$ and $C_p = 4.184 \text{ kJ}/(\text{kg} \cdot \text{K})$:

$$\dot{m}_{cw} = \frac{11169.42}{4.184 \times 10} = 266.96 \text{ kg/s}.$$

8.4 CO₂ emissions

CH₄ → CO₂ is 1:1 molar. Thus $\dot{n}_{CO_2} = \dot{n}_{CH_4} = 0.0154646 \text{ kmol/s}$. Mass flow:

$$\dot{m}_{CO_2} = \dot{n}_{CO_2}(44) = 0.680442 \text{ kg/s}.$$

Annual mass (8000 h/yr):

$$m_{CO_2} = 0.680442 \times (8000 \times 3600) = 1.95967 \times 10^7 \text{ kg/yr}.$$

$$m_{CO_2} = 1.95967 \times 10^{-5} \text{ Gt/yr}.$$

8.5 NO_x emissions

Fuel energy rate in MMBtu/h:

$$13.732560 \text{ MJ/s} \times 3600 = 49437.2 \text{ MJ/h}.$$

Using 1 MMBtu = 1055.06 MJ:

$$\dot{Q}_{fuel} = 46.86 \text{ MMBtu/h}.$$

NO_x mass rate:

$$0.05 \frac{\text{lb}}{\text{MMBtu}} \times 46.86 = 2.343 \text{ lb/h}.$$

Annual:

$$m_{NO_x} = 2.343 \times 8000 \text{ lb} = 18744 \text{ lb} = 8501.7 \text{ kg/yr}.$$

$$m_{NO_x} = 8.50 \times 10^3 \text{ kg/yr}.$$

8.6 Worldwide CO₂ emissions and percent contribution

IEA reports global energy-related CO₂ emissions of 37.4 Gt in 2023.

Source: <https://www.iea.org/reports/co2-emissions-in-2023/emissions-grew-in-2023-but-clean-energy>

Thus:

$$\% = \frac{1.95967 \times 10^{-5}}{37.4} \times 100 = 5.24 \times 10^{-5}\%.$$

9 Lost Work and Second-Law Metrics

$$\dot{W}_{lost} = \sum \dot{n}b_{in} - \sum \dot{n}b_{out} - \dot{W}_s,$$

The condenser and boiler contribute the largest exergy destruction, indicating that the major irreversibilities occur in heat-transfer processes, while the OFWH has negligible lost work due to near-equilibrium mixing.

Table 3: Lost work (exergy destruction) for each unit scaled to a 500 kW net plant output.

Equipment	$\dot{W}_{lost}$ (kW)
Boiler (F-100)	5207.52
HP Turbine (ST-100)	2959.44
LP Turbine (ST-101)	1414.41
Condenser (E-100)	5620.14
Condensate Pump (P-100)	1491.38
Open Feedwater Heater (OFWH)	6.73
Feedwater Pump (P-101)	1959.54
Total	18659.16

10 Optimization of HP Turbine Discharge Pressure (1–10 bar)

Because properties must come from the Excel calculator, the optimization should be performed in Excel by sweeping P_2 from 1 to 10 bar and recomputing:

- HP turbine ideal outlet: enforce $s_{2s} = s_1$ at each P_2 (Goal Seek on x_{2s}) to obtain h_{2s} .
- Real HP outlet: $h_2 = h_1 - \eta_t(h_1 - h_{2s})$.
- OFWH at P_2 : compute $y = (h_8 - h_7)/(h_3 - h_7)$.
- LP turbine: expand from P_2 to P_{cond} to obtain h_5 .
- Net work and duties: $\dot{W}_{net}$, $\dot{Q}_{boiler}$, $\dot{Q}_{cond}$ and then annual costs.

The objective is to minimize annual utilities cost subject to $1 \leq P_2 \leq 10$ bar.

11 Conclusion

Using the CHE Calcuators' properties and a condenser pressure of 0.0738 bar, the cycle was solved and then scaled to deliver 500 kW net power. The required steam rate is 20.8467 MMSCFD. The bleed fraction required for the OFWH condition (implemented via a small liquid enthalpy correction consistent with a 1 C approach) is $y = 0.0018647$. The overall thermal efficiency under the given work/heat assumptions is 4.283%. Utilities usage and emissions were computed directly from the scaled boiler fuel rate and condenser duty.

Table 4: Summary of calculated cycle performance, utilities, and emissions (scaled to 500 kW net output).

Quantity	Calculated Value
Steam flow rate through cycle	20.8467 MMSCFD
Bleed steam mass fraction (y)	0.0018647
Overall thermal efficiency (η_{th})	4.283%
Thermodynamic efficiency (η_{II})	2.61%
Maximum theoretical efficiency (Carnot)	56.7%
Total lost work (all units)	18659.16 kW
Natural gas required flow	1.1164 MMSCFD
Air requirement flow (3% O ₂ dry basis)	12.2186 MMSCFD
Cooling water requirement	266.96 kg/s
CO ₂ emissions per operating year	1.9597×10^{-5} Gt/yr
Worldwide CO ₂ emissions (IEA 2023)	37.4 Gt/yr
Plant contribution to worldwide emissions	5.24×10^{-5} %
NO _x emissions per operating year	8.50×10^3 kg/yr
Annual electricity cost (pumps)	$\$2.95 \times 10^6$ /yr
Annual cooling water cost	$\$1.14 \times 10^5$ /yr
Annual natural gas cost	$\$1.68 \times 10^6$ /yr
Total annual utilities cost (base case, $P_2 = 2$ bar)	$\$4.75 \times 10^6$ /yr
Minimum annual utilities cost	$\$4.32 \times 10^6$ /yr at $P_2 = 4.6$ bar

Statement of AI Tool Use

ChatGPT was used to format and structure this report to optimize readability and organization. AI was used in some aspect to clarify processes to solve, however, all final calculations were done by the student.